

Design and Synthesis of Chlorinated Benzothiadiazole-Based Polymers for Efficient Solar Energy Conversion

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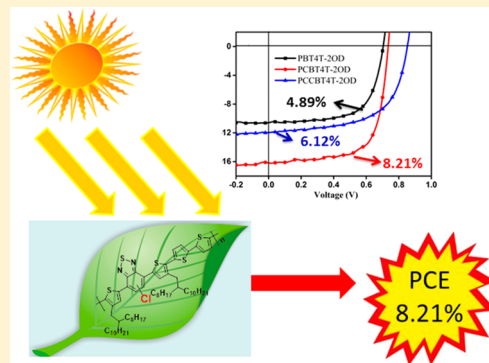
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Supporting Information

ABSTRACT: Chlorinated benzothiadiazole-based polymers with multiple chlorine atoms were designed and synthesized for polymer solar cells to achieve enhanced open-circuit voltage and improved power conversion efficiency. Chlorine substitution was found to affect molecular orientation, increase crystallinity, and thereby alter band gap and charge transport properties. The one-chlorine-substituted PCBT4T-2OD exhibited a larger portion of “face-on” orientation than that of other two polymers, nonchlorinated PBT4T-2OD and two-chlorine-substituted PCCBT4T-2OD, in the polymer:PC₇₁BM blended films. PCBT4T-2OD also showed the largest crystallite sizes in those three polymers. The improved molecular orientation and larger crystallite sizes would definitely facilitate the charge transport in the active layer and enhance the performance of corresponding polymer solar cells. The highest power conversion efficiency of 8.21% with PC₇₁BM was achieved in the photovoltaic device of PCBT4T-2OD. It is approximately 68% higher than that of the nonchlorine analog.



Polymer solar cells (PSCs) have been given significant attention in recent years owing to their benefits of low cost, lightweight, and flexibility,^{1–5} as well as their strong advantages of no need for vacuum processing and high-temperature sintering, nontoxic end products, etc.⁶ Such distinct properties potentially allow for more convenient fabrication methods for PSCs, such as roll-to-roll printing or inkjet printing. Bulk heterojunction (BHJ) PSCs, incorporating solution-processed conjugated polymers and fullerene derivatives, have been widely studied in the past 20 years.^{4,7,8} Enormous research effort has pushed the power conversion efficiency (PCE) of the BHJ PSCs to over 12%.^{9,10} The tremendous boost in the PCE has greatly benefited from the design of new materials and the optimization of device engineering. Using donor–acceptor (D–A) copolymers as the electron donor in PSCs is an important approach to tune the energy levels and optical band gaps.^{5,11,12} In addition, the selection of proper atomic substitution in the backbone is essential and significantly influences the molecular planarity,

intermolecular interactions, charge transport properties, and film morphologies.^{13,14} Over the past decades, investigations into the effects of different atomic substitutions on PSC polymers have attracted much attention.^{13,15} An effective method to finely tune the energy levels and phase separation of the BHJ PSCs is the halogenation of the molecular backbone.¹⁶ For a long time, fluorination has been applied to enrich the multifunctionality of organic solar cells. Fluorine-substituted conjugated polymers tend to lower the highest occupied molecular orbital (HOMO) level of the donor polymer, leading to an increased open-circuit voltage (V_{oc}) and therefore improved device performance.^{6,17–19} However, fluorination usually involves tedious reaction steps with low yield and requires the use of hazardous fluorinating reagents.

Received: January 29, 2017

Accepted: March 7, 2017

Published: March 10, 2017

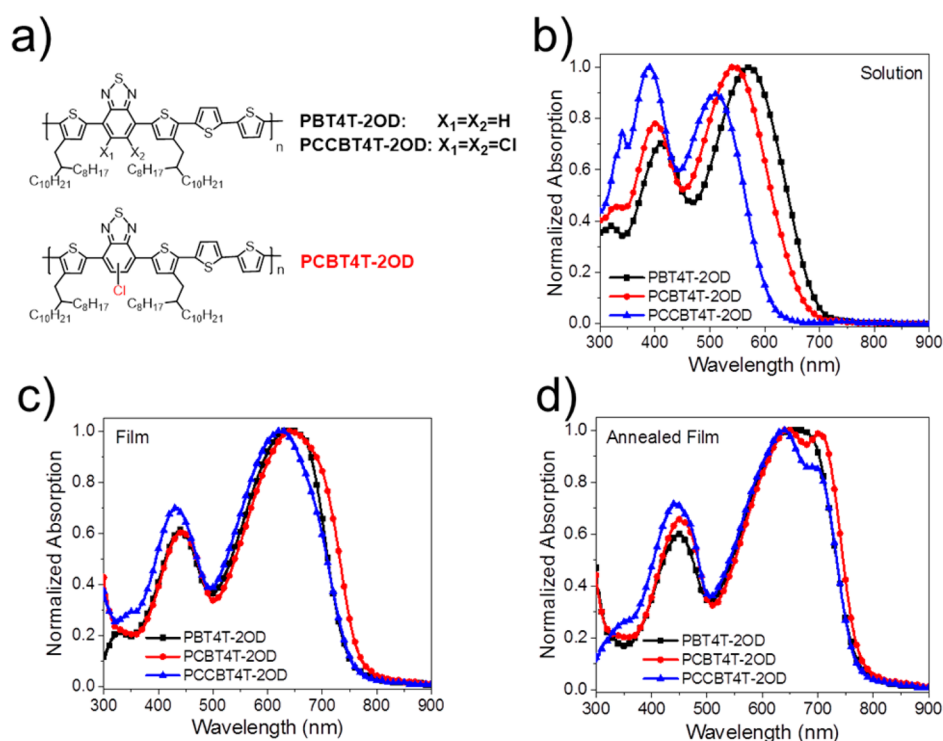


Figure 1. Chemical structures and UV-vis absorption characterization. (a) Chemical structures of the three polymers and UV-vis absorption spectra of the polymers in $CHCl_3$ solution (b), in a thin film (c), and in an annealed film (d).

Table 1. Optical and Electrochemical Properties of the Polymers

polymer	M_n (kDa)	PDI	λ_{max}^{sol} (nm)	λ_{max}^{film} (nm)	E_{opt}^g (eV)	E_{HOMO} (eV)	E_{LUMO} (eV)	E_{CV}^g (eV)
PBT4T-2OD	34.9	1.35	410, 570	439, 647	1.58	−5.12	−3.63	1.49
PCBT4T-2OD	60.7	1.50	401, 540	445, 652	1.59	−5.26	−3.59	1.67
PCCBT4T-2OD	55.6	1.51	389, 511	432, 640	1.61	−5.32	−3.62	1.70

Therefore, the development of an alternative method has become necessary, especially for reel-to-reel solar cell production, which has a large demand for raw materials.¹⁶ So far, chlorination has been less explored as a method to modify polymers for solar device fabrication. Previous studies in small molecules show that the energy levels of chlorinated materials are comparable or even deeper than those of fluorinated materials, because chlorines are able to accommodate more electron density than fluorines.²⁰ Because of the large atomic size, chlorine substitution sometimes induces a contorted π -stacking molecular arrangement. As a result, there are far fewer studies on the properties of chlorinated organic semiconductors in PSCs than those on fluorinated materials, even though chlorination has been shown to be synthetically accessible and have a relatively lower precursor cost.²¹

In this study, conjugated polymers with chlorine atoms substituted on the benzothiadiazole (BT) moiety were synthesized and compared to explore the relationship between the number of chlorine atoms and the photovoltaic properties. It was found that polymers with chlorine atom substitution showed dramatic improvements in device performance, especially for V_{oc} and short-circuit current (J_{sc}). PSCs based on polymers with one chlorine atom substituted on the BT moiety delivered a PCE of up to 8.21% with PC₇₁BM, which is much higher than the best value (4.89%) from its nonchlorine analog.

Synthetic routes for all monomers and copolymers are shown in the Supporting Information (Scheme S1). The parent polymer PBT4T-2OD was synthesized for comparison.²² PCBT4T-2OD and PCCBT4T-2OD (Figure 1a) were synthesized by Stille coupling using $Pd_2(dba)_3$ and $P(o\text{-tolyl})_3$ in chlorobenzene solvent. The molecular weights of the polymers were determined by gel permeation chromatography using THF as the solvent. The number-average molecular weights (M_n) of PCBT4T-2OD and PCCBT4T-2OD were determined to be 60.7 and 55.6 kDa, with polydispersity indices (PDI) of 1.50 and 1.51, respectively (Table 1). The two chlorinated polymers were prepared with comparable M_n and PDI values by adjusting the synthetic conditions to minimize the effect of molecular weight on the various properties of the polymers. In the fabrication of solution-processed solar cells, the thermal stability of the donor material is of great importance. Thus, thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) was used to evaluate the thermal properties of the resulting copolymers (Figures S1 and S2). Under N_2 atmosphere, the onset temperature of 5% weight loss was approximately 422 °C for PBT4T-2OD, 430 °C for PCBT4T-2OD, and 405 °C for PCCBT4T-2OD, which indicated that these polymers were thermally stable enough for the study of their PSCs. The DSC scans of the polymers are shown in Figure S2. Each polymer exhibited an endothermic peak upon heating and an exothermic peak upon cooling, indicating the melting and crystallization transitions of the

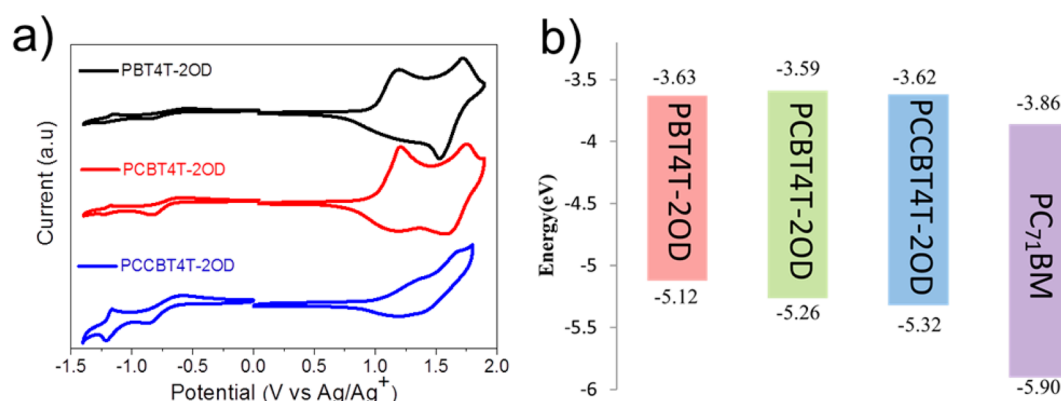


Figure 2. (a) Cyclic voltammograms of polymers in thin films drop-casted on glassy carbon electrodes at a scan rate of 100 mV s⁻¹ and (b) energy level diagrams for the polymers and PC₇₁BM.

Table 2. Photovoltaic Properties of the Polymer-Based Devices

polymer	V _{OC} (V)	J _{SC} (J _{cal}) ^a (mA cm ⁻²)	FF (%)	PCE (avg) ^b (%)	μ _n (cm ² V ⁻¹ s ⁻¹)
PBT4T-2OD	0.70	10.58 (10.23)	65.45	4.89 (4.57 ± 0.14)	7.6 × 10 ⁻⁵
PCBT4T-2OD	0.73	16.18 (15.67)	68.97	8.21 (8.03 ± 0.04)	5.1 × 10 ⁻⁴
PCCBT4T-2OD	0.85	11.93 (11.52)	60.14	6.12 (6.02 ± 0.08)	2.2 × 10 ⁻⁴

^aThe integrated J_{cal} was calculated from the EQE response spectra. ^bAverage value of PCE ± standard deviation for 20 different devices.

polymers, respectively. PCBT4T-2OD showed a melting point of 205 °C, which was higher than that of PBT4T-2OD (176 °C) and PCCBT4T-2OD (168 °C) and could be attributed to its better packing of the polymer chains and stronger intermolecular interactions due to the monochlorine substitution at the BT unit.

The ultraviolet–visible (UV–vis) absorption spectra of the polymers were measured in solution, in films (Figure 1b,c), and in annealed films (150 °C for 10 min) (Figure 1d). Detailed data of the optical properties are summarized in Table 1. In chloroform solution, the three polymers presented two strong absorption bands in the ranges of 350–500 and 500–700 nm. The addition of chlorine atoms to the BT unit in PCBT4T-2OD and PCCBT4T-2OD led to a blue-shift due to enhanced steric hindrance from the chlorine atoms. PBT4T-2OD had an absorption maximum at 570 nm in solution and 647 nm in the thin film, with a red-shift of 77 nm from the solution to the film. The addition of one chlorine atom to the BT unit in PCBT4T-2OD led to a slight red-shift (652 nm) compared to PBT4T-2OD in thin-film form. In addition, from solution to film, this polymer showed a red-shift in the maximum absorption of 112 nm, which suggests that strong aggregations formed in the solid state even with the addition of a large chlorine atom in the PCBT4T-2OD film. However, the addition of two chlorine atoms to the BT unit in PCCBT4T-2OD led to a blue-shift both in solution (511 nm) and in the thin film (640 nm), in comparison to the other two polymers, which indicates that the two chlorine atoms together reduce π – π stacking in polymer films because of their strong steric hindrance. After the thin films were annealed at 150 °C for 10 min, a red-shift in the maximum absorption peak (656, 653, and 642 nm) was observed for all three polymers. In addition, a strong shoulder appeared at 705 nm in the PCBT4T-2OD spectra, and a relatively weak shoulder appeared at 697 nm in the PCCBT4T-2OD spectra, which shows that thermal annealing helped the chlorinated polymers form a better morphology with stronger π – π intermolecular interactions. The optical band gap calculated from the wavelength of

absorption onset was 1.58 eV for PBT4T-2OD, which was smaller than that of PCBT4T-2OD (1.59 eV) and that of PCCBT4T-2OD (1.61 eV). These results indicated that chlorine atoms substituted on the BT moieties would greatly influence the aggregation of the polymers in the film state, especially with proper thermal annealing. With the addition of one chlorine atom, PCBT4T-2OD showed the largest red-shift in films with the strongest π – π stacking, which would benefit charge transfer in polymer films and increase the PCE of photovoltaic devices.

The electrochemical properties of the polymers were evaluated by cyclic voltammetry (CV). The oxidative peaks of these polymers were 1 order of magnitude stronger than the reductive peaks (Figure 2a). The estimated energy levels of the polymers are plotted in Figure 2b, and the electronic properties of these three polymers are summarized in Table 1. The lowest unoccupied molecular orbital (LUMO) levels of PBT4T-2OD, PCBT4T-2OD, and PCCBT4T-2OD were estimated to be -3.63, -3.59, and -3.62 eV, respectively. However, the HOMO levels of PCBT4T-2OD and PCCBT4T-2OD were -5.26 and -5.32 eV, which were both deeper than that of PBT4T-2OD (-5.12 eV) by 0.14 and 0.20 eV, respectively. The HOMO levels were therefore reduced with increasing numbers of chlorine atoms. PCBT4T-2OD may form moderate aggregations in the solid state, while PCCBT4T-2OD showed less aggregation, in accordance with the results obtained from the optical characterizations. The frontier orbital density distributions for these polymers are shown in Figure S2. It was not surprising that modification of the BT unit with chlorine atoms predominantly affects the energy of the HOMO level because the wave function of the HOMO level, determined from theoretical calculations, was concentrated on the thiophene unit.

To evaluate the potential of these polymers as donor materials and understand the relationship between the structure, properties, and performance, BHJ solar cells were fabricated with an inverted device structure, ITO/ZnO/polymer:PC₇₁BM/MoO₃/Ag. The D–A blend ratio for all

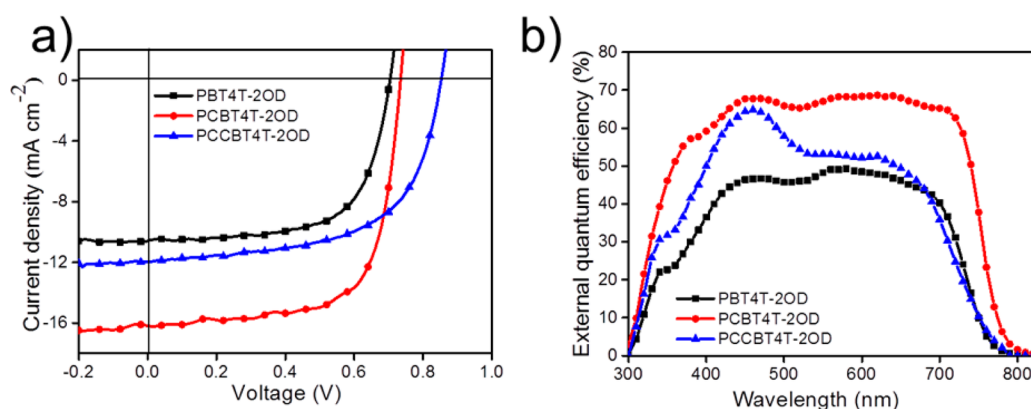


Figure 3. (a) Current density–voltage (J – V) characteristic and (b) external quantum efficiency (EQE) spectra of the BHJ PSCs with 1:1.2 w/w blends of PBT4T-2OD:PC₇₁BM, PCBT4T-2OD:PC₇₁BM, and PCCBT4T-2OD:PC₇₁BM in CB with 3% DIO.

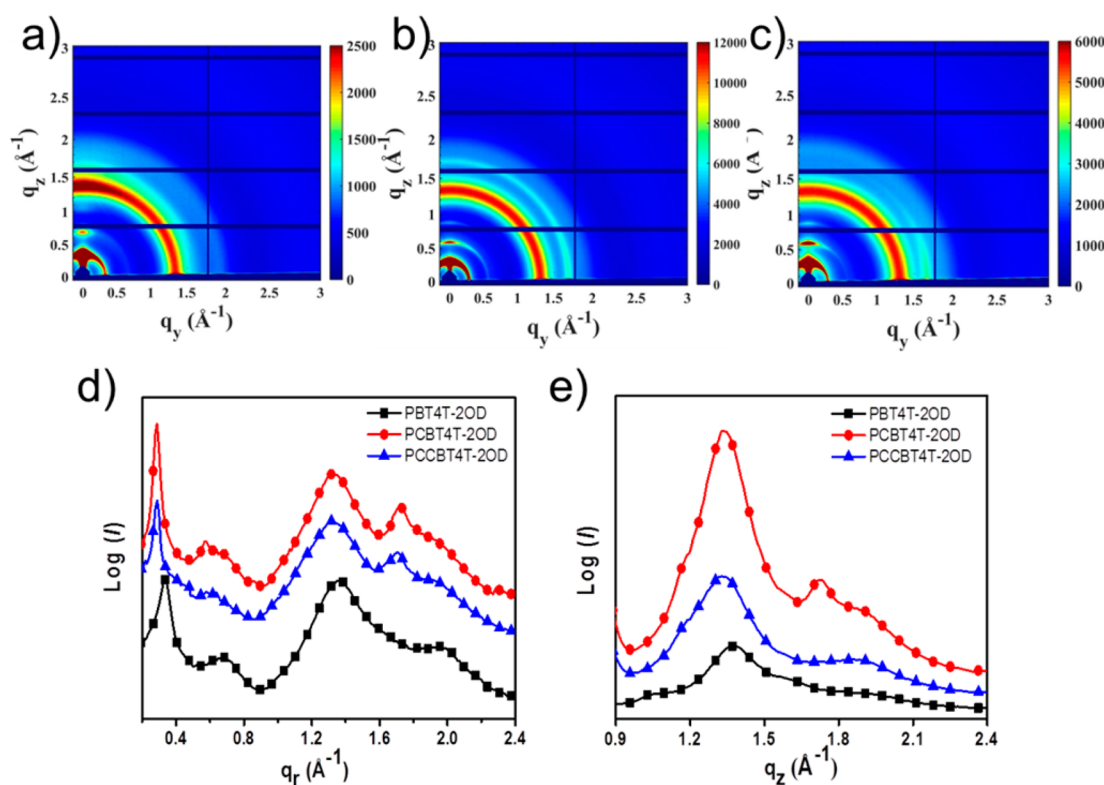


Figure 4. (a–c) 2D GIWAXS patterns of the PBT4T-2OD:PC₇₁BM, PCBT4T-2OD:PC₇₁BM, and PCCBT4T-2OD:PC₇₁BM blend films. Line-cut in the (d) in-plane and (e) out-of-plane direction.

polymers was 1:1.2 (polymer:PC₇₁BM) by weight at a total solids concentration of 20 mg mL⁻¹ in chlorobenzene containing 3% 1,8-diiodooctane (DIO) as an additive. The photovoltaic performances of PSCs based on these polymers were systematically evaluated, and the optimal photovoltaic parameters of the devices are listed in Table 2. Characteristic current density–voltage (J – V) curves for these optimal devices are shown in Figure 3a. Among these polymers, PCBT4T-2OD showed the highest PCE of 8.21% with the highest J_{sc} of 16.18 mA cm⁻² and an enhanced V_{oc} of 0.73 V, compared with those of the chlorine-free polymer PBT4T-2OD. Most impressively, the V_{oc} of the devices based on these polymers increased monotonically with the number of chlorine atoms on the BT moieties. As expected, the HOMO levels became deeper when more chlorine atoms were added onto the BT units; the V_{oc} of the devices of the three polymers also become larger with more

chlorine atoms in the polymer backbone. PCBT4T-2OD exhibited a V_{oc} (0.73 V) higher than that of PBT4T-2OD (0.70 V), as its HOMO level was lowered by the addition of one chlorine atom substituted on the BT moieties. PCCBT4T-2OD, with two chlorine atoms attached to the backbone of the polymer, had the highest V_{oc} (0.85 V) of these polymers. The J_{sc} values were also closely correlated with the substitution of chlorine atoms onto the BT units in these polymers, and the lowered V_{oc} value for the polymer without chlorine substitution produced a subnormal PCE of 4.89%. The monochloride PCBT4T-2OD led to an obviously increased J_{sc} up to 16.18 mA cm⁻². However, the introduction of two chlorine atoms onto the BT unit in PCCBT4T-2OD led to a large drop in the J_{sc} compared to that of PCBT4T-2OD, which had the highest V_{oc} of the three polymers, resulting in a PCE of 6.12%. The substitution of two chlorine atoms onto the BT unit may

decrease intermolecular interactions and influence the morphology of the polymer blend films. The balanced V_{oc} and J_{sc} of monochlorinated **PCBT4T-2OD** achieved the highest PCE of 8.21% in this series of polymers. The highest J_{sc} exhibited by **PCBT4T-2OD** was likely to be, in part, due to the optimization of the LUMO energy level, which led to a 0.18 eV increase in the band gap of **PCBT4T-2OD** compared to that of **PBT4T-2OD**, which would benefit efficient charge transfer in the blend films. The hole mobility of **PCBT4T-2OD**, measured by the space-charge-limited current (SCLC) method with hole-only device structure of ITO/PEDOT:PSS/polymer:PC₇₁BM/MoO₃/Ag, was also an order magnitude higher than that of **PBT4T-2OD** and **PCCBT4T-2OD** (seen in Table 2). These observations also help to explain the higher current measured in the devices made from **PCBT4T-2OD**. According to the device EQE spectrum (Figure 3b), the EQE profile of **PCBT4T-2OD** showed values from 300 to 800 nm that were much higher than those of the other two polymers; in particular, much higher values of approximately 70% were maintained from 450 to 750 nm, which was consistent with the film absorption profile of the annealed polymers, shown in Figure 1d, and the large enhancement in J_{sc} . The maximum EQE of **PCCBT4T-2OD** was 65% at 460 nm, and the wavelength range of 500–700 nm showed EQEs under 50%. Moreover, the maximum EQE of **PBT4T-2OD** was much lower than that of **PCBT4T-2OD**, which produced a decrease in the J_{sc} value of its corresponding device. The calculated integrated current densities from the EQE curves matched well with the J_{sc} values from the J – V curves. The fullerene excitons in this blend contributed significantly more to the photocurrent than the polymer excitons. The negligible hole mobility (Figure S3) of **PCCBT4T-2OD** also contributed to the low J_{sc} value, even if excitons were able to separate despite the low hole mobility, possibly resulting in a buildup of charges that would increase nongeminate recombination losses.

The optimal morphology of the active layer is critically important for the performance of PSCs. It is believed that chlorine atoms could impact the miscibility and solubility of polymers blended with PC₇₁BM. The film morphology of the blend films should also be influenced by the number of chlorine atoms. To gain an in-depth understanding of the dramatic enhancement in charge transport and the properties associated with chlorination, the polymer:PC₇₁BM blend films were investigated using grazing-incidence wide-angle X-ray scattering (GIWAXS) (Figure 4). The films were prepared under the conditions used to fabricate the optimized devices. The 2D GIWAXS patterns in Figure 4a–c showed that these three polymers had dominant “edge-on” conformations in the polymer:PC₇₁BM blend films, and their corresponding in-plane and out-of-plane line-cuts (Figure 4d and 4e) exhibited a scattering peak at $\sim 1.7 \text{ \AA}^{-1}$, arising from the π – π stacking of the polymer backbones with a distance of 3.7 Å. However, **PBT4T-2OD**:PC₇₁BM had a weak π – π stacking scattering; **PCBT4T-2OD**:PC₇₁BM exhibited a ring-like π – π stacking scattering with a relatively weak intensity along the z -axis; **PCCBT4T-2OD**:PC₇₁BM showed the scattering only in the in-plane direction. This demonstrated that the different number of chlorine atoms substituted on the BT moieties in each repeating unit could influence the orientation of the polymer chains. The one-chlorine-substituted polymer **PCBT4T-2OD** had a larger portion of the “face-on” conformation in the blended films than the two-chlorine-substituted polymer **PCCBT4T-2OD**. It was accepted that the mixed “edge-on”

and “face-on” conformation of the **PCBT4T-2OD** polymer in the active layer could be beneficial for promoting vertical charge transport between the electrodes. In addition, the full width at half-maximum (fwhm) of scattering peaks at $\sim 0.3 \text{ \AA}^{-1}$ ($\text{fwhm}_{\text{PBT4T-2OD}} = 0.084 \text{ \AA}^{-1}$, $\text{fwhm}_{\text{PCBT4T-2OD}} = 0.034 \text{ \AA}^{-1}$, and $\text{fwhm}_{\text{PCCBT4T-2OD}} = 0.038 \text{ \AA}^{-1}$, respectively) indicated that the crystallite sizes of these polymers were in the sequence **PBT4T-2OD** < **PCCBT4T-2OD** < **PCBT4T-2OD**. The larger crystallite size was able to facilitate charge transport, consistent with the measurements on the hole mobility of these polymers (Table 2). Consequently, incorporating the proper number of chlorine atoms on the BT units in the polymers could fine-tune the local ordering like π – π molecular orientation and crystallinity, thereby facilitating charge transport in the active layer and improving the performance of polymer solar cells.

Additionally, the morphology of the three blend films was studied by tapping-mode atomic force microscopy (AFM) to further understanding the effect of chlorine substitution. As shown in Figure S4, the **PCBT4T-2OD**:PC₇₁BM blend film showed a smoother morphology with a root-mean-square roughness (RMS) of 1.95 nm, compared with that of **PBT4T-2OD**:PC₇₁BM (RMS = 6.83 nm) and **PCCBT4T-2OD**:PC₇₁BM (RMS = 3.87 nm). These data showed that **PCBT4T-2OD** maintained reasonably small polymer domains and exhibited good intermixing with PC₇₁BM, which allowed **PCBT4T-2OD** to demonstrate the highest photovoltaic performance among these three polymers. The partial “face-on” molecular orientation also provides a direction to optimize this system for better solar energy conversion.

In summary, we have designed and synthesized two conjugated polymers (**PCBT4T-2OD** and **PCCBT4T-2OD**) containing different numbers of chlorine atoms substituted on the BT moieties. Chlorine substitution was found to significantly alter the band gap of the polymers and fine-tune the local ordering in polymer blend films, such as π – π molecular orientation and crystallinity. That would facilitate the charge transport in the active layer and improve the performance of polymer solar cells. PSCs with **PCBT4T-2OD**:PC₇₁BM blend film as the active layer showed a PCE of 8.21% with a V_{oc} of 0.73 V, a J_{sc} of 16.18 mA cm^{-2} , and a fill factor (FF) of 68.97%. In comparison with the chlorine-free polymer **PBT4T-2OD**, the chlorine-substituted polymers **PCBT4T-2OD** and **PCCBT4T-2OD** possessed deeper HOMO energy levels, formed reasonably small polymer domains, and exhibited good intermixing with PC₇₁BM. Both chlorine-substituted polymers showed superior solar energy conversion properties compared with their nonchlorine analog, especially the PCE of **PCBT4T-2OD** was approximately 68% higher than that of **PBT4T-2OD**. Our results demonstrated that chlorine substitution can be an effective alternative to modulate properties and promote the performance of polymer solar cells.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsenerylett.7b00092.

Synthetic procedures; TGA, DSC, and theoretical calculations data; AFM images (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was financially supported by the National Basic Research Program of China (2013CB834805), the Shenzhen Fundamental Research program (JCYJ20150630145302237, JCYJ20160504151731734), the Shenzhen Key Lab (ZDSYS201505291525382), and the Shenzhen Peacock program (KQTD20140630110339343). W.C. gratefully acknowledges financial support from the U.S. Department of Energy, Office of Science, Materials Sciences and Engineering Division. Use of the Advanced Photon Source (APS) at Argonne National Laboratory was supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences, under Contract No. DE-AC02-06CH11357.

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